

Machine Learning Assignment - 2

M.Tech DSE - Classification Models and Streamlit Deployment

Submission Links

GitHub Repository Link: https://github.com/syayounus/ML_assignment2.git

Live Streamlit App Link: <https://mlassignment2-deofcsapppfqtpv6lctsvjp.streamlit.app/>

Problem Statement

This project implements and compares multiple classification models on a public dataset. It also provides an interactive Streamlit app for model selection, test CSV upload, metric display, confusion matrix, classification report, and prediction preview.

Dataset Description

The Breast Cancer Wisconsin Diagnostic dataset is used. It is a binary classification dataset with 569 instances and 30 numerical features. The target column is target, where 0 means malignant and 1 means benign.

Model Comparison Table

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9860	0.9977	0.9889	0.9889	0.9889	0.9700
Decision Tree	0.9371	0.9186	0.9551	0.9444	0.9497	0.8657
kNN	0.9790	0.9845	0.9677	1.0000	0.9836	0.9555
Naive Bayes	0.9371	0.9893	0.9263	0.9778	0.9514	0.8650
Random Forest (Ensemble)	0.9580	0.9949	0.9565	0.9778	0.9670	0.9098

Observations

ML Model Name	Observation about model performance
Logistic Regression	Performed very strongly because the dataset has mostly numerical features with clear linear separation after scaling.
Decision Tree	Good and easy to interpret, but slightly more sensitive to train/test split and can overfit if depth is not controlled.
kNN	Performed well after feature scaling because distance-based comparison works better when all features are on comparable scales.

Machine Learning Assignment - 2

Naive Bayes	Fast and simple baseline. It performed reasonably but can be weaker when feature independence assumptions are not fully true.
Random Forest (Ensemble)	Strong overall performance because it combines many decision trees and reduces single-tree overfitting.
Overall Winner	Logistic Regression, based on the strongest combination of F1, MCC, and AUC.

GitHub README Content

Machine Learning Assignment 2 - Classification Models and Streamlit Deployment

a. Problem Statement

The objective of this assignment is to build and compare multiple supervised classification models on one public classification dataset. The trained models are exposed through an interactive Streamlit web application where users can upload test data, select a model, and view evaluation results.

b. Dataset Description

Dataset selected: Breast Cancer Wisconsin Diagnostic dataset from the UCI Machine Learning Repository.

- Task type: Binary classification
- Number of instances: 569
- Number of input features: 30
- Target classes: malignant and benign
- Minimum assignment requirement satisfied: more than 500 instances and more than 12 features
- Source: UCI Machine Learning Repository / sklearn built-in dataset loader

The target column used in this project is target.

- 0 = malignant
- 1 = benign

c. GitHub Repository Link

https://github.com/syayounus/ML_assignment2.git

d. Live Streamlit App Link

<https://mlassignment2-deofcsapppfqtvp6lctsvjp.streamlit.app/>

e. Models Used and Evaluation Metrics

All models were trained and evaluated on the same train-test split with random_state=42.

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9860	0.9977	0.9889	0.9889	0.9889	0.9700
Decision Tree	0.9371	0.9186	0.9551	0.9444	0.9497	0.8657
kNN	0.9790	0.9845	0.9677	1.0000	0.9836	0.9555
Naive Bayes	0.9371	0.9893	0.9263	0.9778	0.9514	0.8650
Random Forest (Ensemble)	0.9580	0.9949	0.9565	0.9778	0.9670	0.9098

f. Observations on Model Performance

ML Model Name	Observation about model performance
Logistic Regression	Performed very strongly because the dataset has mostly numerical features with clear linear separation after scaling.
Decision Tree	Good and easy to interpret, but slightly more sensitive to train/test split and can overfit if depth is not controlled.
kNN	Performed well after feature scaling because distance-based comparison works better when all features are on comparable scales.
Naive Bayes	Fast and simple baseline. It performed reasonably but can be weaker when feature independence assumptions are not fully true.
Random Forest (Ensemble)	Strong overall performance because it combines many decision trees and reduces single-tree overfitting.
Overall Winner	Logistic Regression, based on the strongest combination of F1, MCC, and AUC.

g. Streamlit App Features

The app provides the following required features:

- CSV test data upload option
- Model selection dropdown
- Evaluation metrics display
- Confusion matrix display
- Classification report display
- Prediction output preview

h. Repository Structure

```

project-folder/
|-- app.py
|-- requirements.txt
|-- README.md
|-- test_data.csv
|-- train_data.csv
|-- model_metrics.csv
|-- model/
|   |-- train_models.py
|   |-- logistic_regression.pkl
|   |-- decision_tree.pkl
|   |-- knn.pkl
|   |-- naive_bayes.pkl
|   |-- random_forest_ensemble.pkl
|   |-- feature_columns.json
|   |-- target_names.json
|   |-- evaluation_reports.json

```

i. How to Run Locally

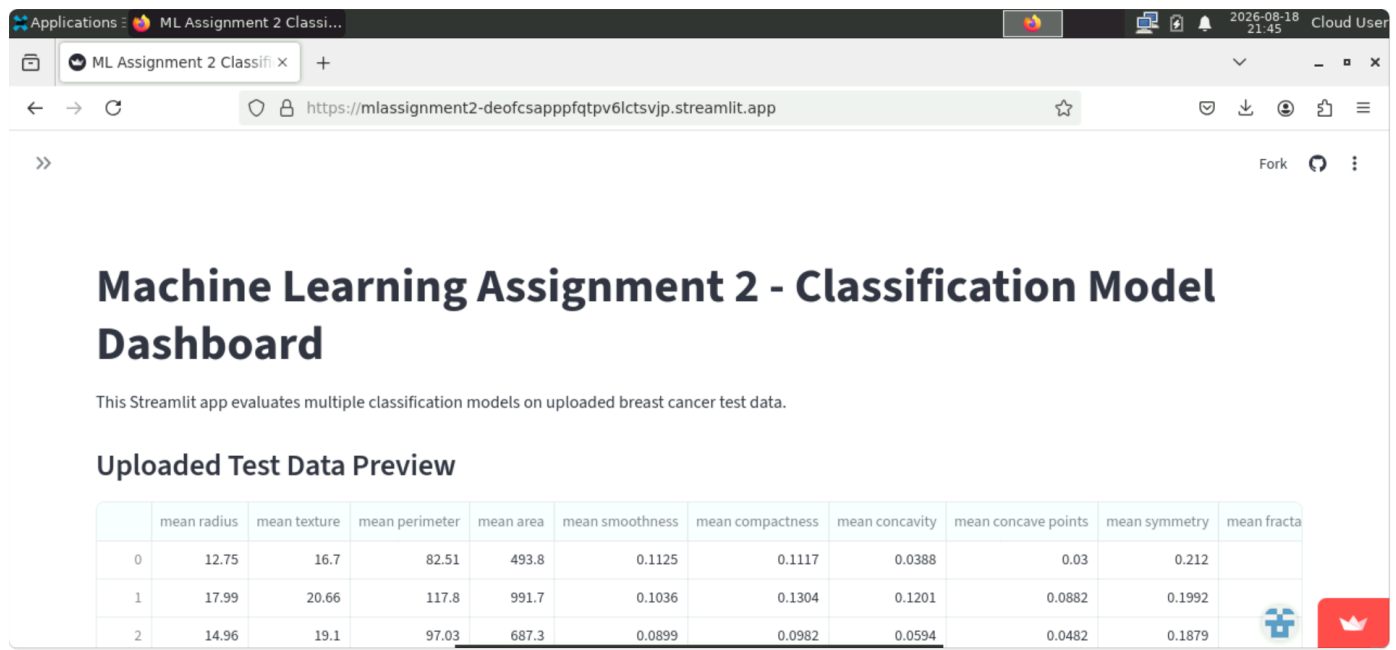
```
pip install -r requirements.txt
streamlit run app.py
```

Then upload test_data.csv, select any model from the dropdown, and review the metrics.

j. Deployment Steps

1. Upload this project folder to a new GitHub repository.
2. Go to Streamlit Community Cloud.
3. Click New App.
4. Select the GitHub repository and branch.
5. Set the main file path as app.py.
6. Click Deploy.
7. Copy the deployed app URL and replace the placeholder in this README and in the final PDF.

k. BITS Virtual Lab Screenshot



Machine Learning Assignment 2 - Classification Model Dashboard

This Streamlit app evaluates multiple classification models on uploaded breast cancer test data.

Uploaded Test Data Preview

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fracta
0	12.75	16.7	82.51	493.8	0.1125	0.1117	0.0388	0.03	0.212	
1	17.99	20.66	117.8	991.7	0.1036	0.1304	0.1201	0.0882	0.1992	
2	14.96	19.1	97.03	687.3	0.0899	0.0982	0.0594	0.0482	0.1879	